

OONI: Open Observatory of Network Interference



Applying Machine Learning to Global Network Telemetry

SDC486L: Advanced Data Analytics and AI/ML | Capstone Project

THE DATASET

Open Observatory of Network Interference (OONI)

Free, publicly available global measurement network hosted on AWS S3



240+

Countries in dataset

5,000

Records in project sample

30 mo

Jan 2022 to Jun 2024

20

Countries analyzed

Why OONI?

The same primitives OONI uses to detect censorship (DNS mismatch, TCP latency, HTTP header flags) are the same signals network operations teams monitor every day. This dataset maps directly to real operational problems in global network infrastructure.

What it captures

Website reachability tests, DNS consistency checks, HTTP header manipulation detection, TCP connectivity probes; run by volunteer probes across 240+ countries, updated hourly since 2012.



Three Analytical Tasks

Each maps directly to a real problem in network operations and AIOps

01

Classification (Task 1)

Can a ML model distinguish confirmed censorship from measurement noise?

Ops parallel: Separating real signal from false positives in a NOC.

02

Clustering (Task 2)

Do countries cluster by blocking behavior without political labels?

Ops parallel: Grouping devices by behavioral fingerprint for policy automation.

03

Trend Prediction (Task 3)

Can leading indicators predict censorship escalation 30 days out?

Ops parallel: Predictive alerting before an incident peaks.

Key Findings Before Modeling

`dns_mismatch_score` ($r=0.61$) is the dominant censorship signal — clusters align with Freedom House tiers without political labels



Class Imbalance

6.6:1 ratio between clean (ok) and confirmed censorship events.
Required stratified sampling. Overall accuracy alone is misleading.

Missing Data

3% missing in `dns_mismatch_score` and `tcp_connect_ms`. Addressed via median imputation to preserve majority-class central tendency.

Geographic Range

Blocking rates: 3% (US/Germany) to 71% (China/Iran). Natural bimodal structure supported the three-cluster hypothesis.

Country Anomaly Rate by Cluster

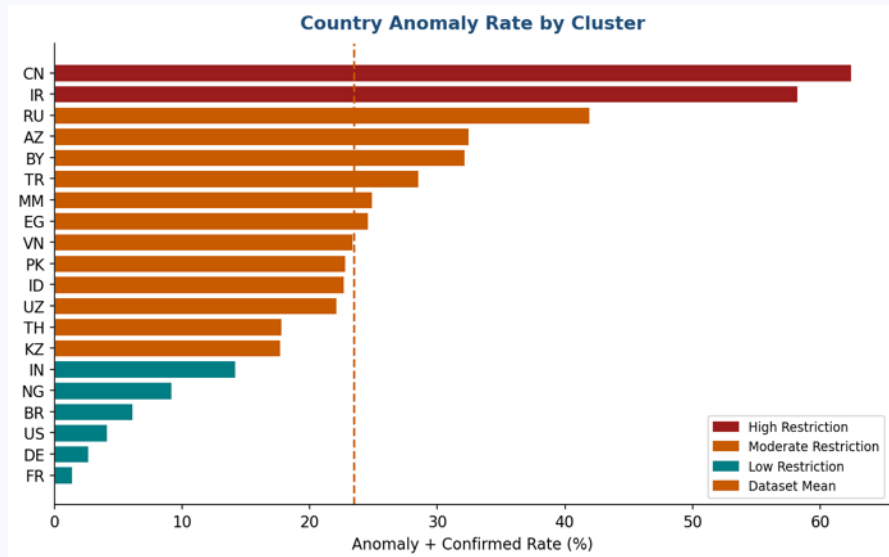


Chart from project analysis — 20 countries, colored by unsupervised cluster assignment

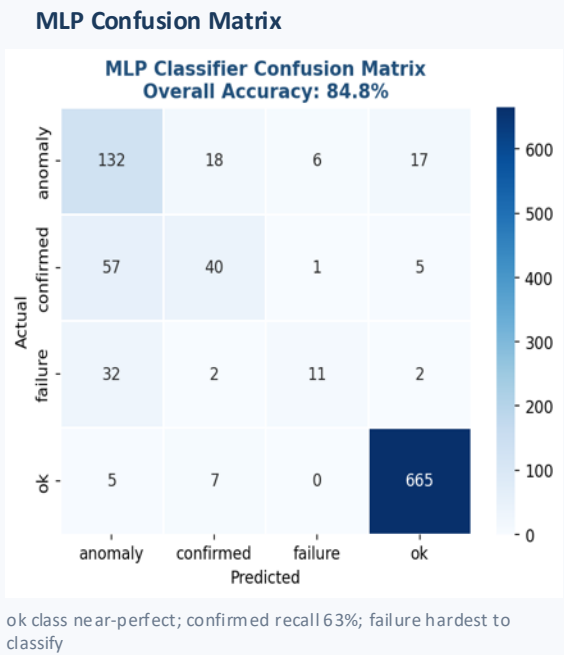
Baseline vs Neural Network — All Three Tasks

Honest assessment: overall accuracy masks important per-class variation



Task	Baseline	Neural Network
Classification	LR: 84.8%	MLP: 86.5%
Clustering	K-Means: 0.40	Autoencoder: 0.56 (+40%)
Trend Pred.	ARIMA MAE: 0.028	LSTM MAE: 0.029

Key: Classification is largely linearly separable. Clustering shows largest NN gain (+40% silhouette). Trend prediction limited by 30-month series length.





Three Operationally Relevant Scenarios

Testing model behavior under real-world network engineering conditions

S1

Censorship Fingerprint

200 synthetic records with strong censorship signal (dns_mismatch 0.85-0.99, resolver_match=0, header_mismatch=1). Result: 89.5% combined detection. Confirmed rate 3.7x baseline. Model is production-viable.

S2

Platform Data Throttling

Elastic-style pre-aggregation: dns_mismatch capped at 0.30, tcp_ms flattened, header_mismatch zeroed. Result: 8.8pp accuracy drop. 24.3pp confirmed recall drop. 1 in 4 real events missed.

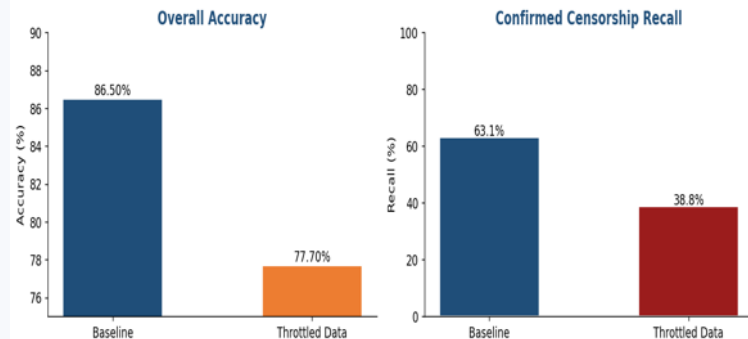
S3

Cross-ASN Diversity Shift

Shifted n_asns +/- 1.5 std devs to simulate ISP consolidation and expansion. Result: 0/20 countries changed cluster. Model is behavior-driven, robust across network topologies.

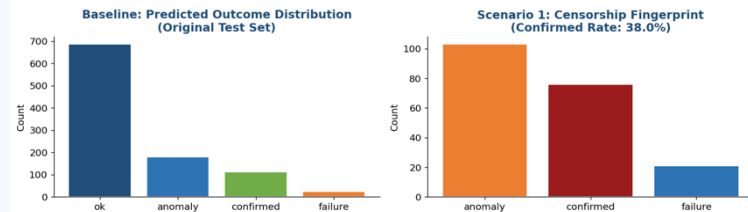
Scenario 2: Data Throttling Impact

Figure S2: Scenario 2 — Impact of Platform Data Throttling on Model Performance



Scenario 1: Censorship Detection

Figure S1: Scenario 1 — Confirmed Censorship Fingerprint vs Baseline





What the Data Tells Us to Do

Actionable findings for network engineering and operations teams

01

Do not pre-aggregate telemetry before it reaches the model

Scenario 2 quantifies the cost at 24.3pp confirmed recall loss. For AIOps platforms on Elasticsearch backends hitting retention limits, this is the data-driven argument for migrating to a columnar store like ClickHouse that handles high-cardinality telemetry without cardinality penalties.

02

Prioritize `dns_mismatch_score` and `resolver_match` as primary alert triggers

These two features drive detection capability more than any others. Any monitoring pipeline should collect and retain them at full resolution. Pre-aggregating them first causes disproportionate harm relative to other feature truncations.

03

Apply behavioral clustering to internal network telemetry

The autoencoder clustering approach can group devices, sites, or ASNs by behavioral fingerprint without manual rule authoring. Cluster stability under ASN shifts confirms the model generalizes across different network topologies.



The Broader Point

The skills that make someone good at network troubleshooting transfer directly to applied machine learning:

Anomaly detection, feature engineering on time-series telemetry, separating signal from noise at scale. This project is proof of concept for that transfer.

Dataset

OONI Open Data on AWS | ooni.org/data/

Research

Azzabi et al. (2024) | Brown & Schreiber (2025) | Thalpati & Ramalingam (2023)

GitHub

<https://github.com/thepacketpoet-school/sdc486l>